

Token identity provides a fixed routing signal for residual experts in language models

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Abstract

Mixture-of-experts language models usually learn routers from contextual hidden states. We test a feed-forward routing primitive in which token identity allocates narrow residual expert capacity while a shared dense multilayer perceptron preserves contextual processing. In a matched single-seed comparison, 306.5-million-parameter dense and token-routed models were trained on 8 billion FineWeb-Edu tokens. The token-routed model reached an evaluation-stream negative log-likelihood of 2.9329, compared with 2.9482 for dense, but trained more slowly. Final checkpoints were near parity on zero-shot ARC-Easy, PIQA and HellaSwag, with word perplexity 35.20 versus 35.79 on WikiText-2. A separate learned-router diagnostic reversed its short-budget advantage over dense at a matched 1-billion-token budget. These results identify fixed token identity as a useful architectural primitive in this setting, without establishing general superiority over dense or learned routing. Larger-scale, multi-seed validation remains necessary.

Mixture-of-experts (MoE) language models conditionally select parameters for each input. Their routers commonly score experts from contextual hidden states and use auxiliary losses or bias updates to prevent load collapse[1–3]. This design couples two questions: whether conditional capacity is useful and whether selecting that capacity requires a learned contextual decision. Separating them can clarify what expert routing contributes.

Fixed assignment is a useful counterpoint. Hash Layers showed that hashes of local token features can compete with learned routers in sparse language models[4]. Token identity is an especially austere routing signal: it cannot distinguish different uses of the same token. Yet it is stable, free of router parameters and may repeatedly expose lexical items to a consistent parameter subspace. If a shared path retains common contextual computation, such a fixed signal might still allocate useful residual capacity.

We test that possibility in a deliberately asymmetric architecture. Every token passes through a shared SwiGLU multilayer perceptron (MLP). Its token identifier selects two additional narrow experts, which receive the same contextual hidden state and add a gated residual. Thus token identity selects functions but does not replace contextual representations. We evaluate this primitive inside a decoder-only transformer so that the routing effect is isolated from changes to sequence modelling. The separation between a shared contextual substrate and fixed residual access is not conceptually tied to attention, but the present experiments do not test backbone independence. The primary experiment compares this architecture with a parameter-matched dense model at 306.5 million parameters and 8 billion training tokens. We also evaluate the final checkpoints on four standard tasks and report separate 100-million-parameter controls with fixed, dense and learned contextual routing.

The token-routed model crosses the dense training curve after approximately 0.8 billion tokens and retains a small negative-log-likelihood (NLL) advantage at the final common evaluation. It is slower in the present implementation, and task accuracy differences are smaller than one reported standard error. A learned contextual router leads dense at short budget but trails it at a matched

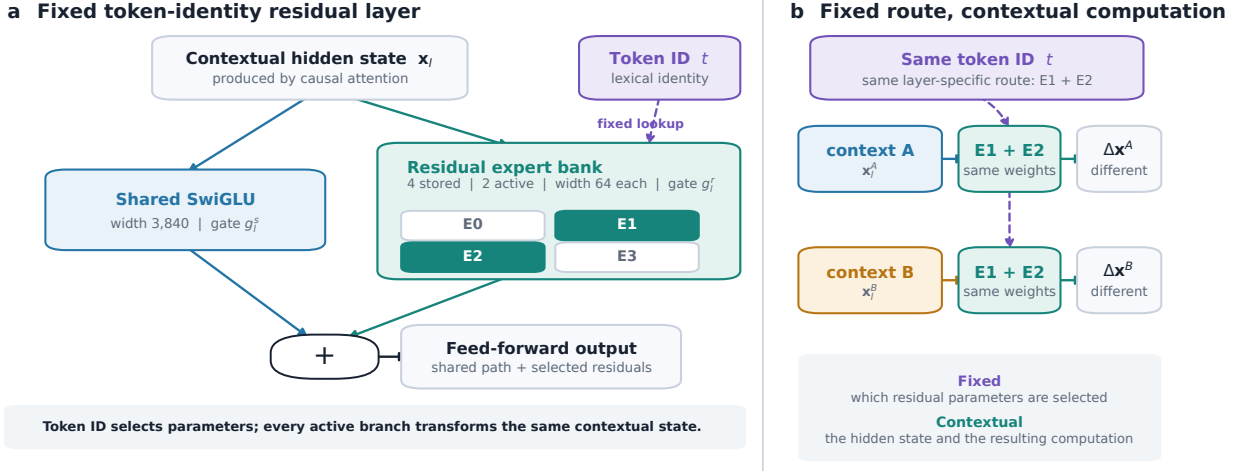


Figure 1: **Fixed token-identity routing controls residual parameter selection.** **a**, Attention produces a contextual hidden state $\mathbf{x}$. Every token traverses the shared SwiGLU branch, while its token identifier selects two narrow residual experts through a fixed layer-specific table. **b**, Different contextual states associated with the same token select the same expert pair but produce different residual outputs. Token identity therefore fixes parameter selection, not the contextual computation performed by those parameters.

1-billion-token budget. The evidence therefore supports a narrow conclusion—fixed token identity can allocate useful residual capacity in this setting—rather than a general ranking of routing mechanisms.

Results

Token identity selects a narrow residual path

The architecture changes only the transformer’s feed-forward sublayer (Fig. 1). Attention remains causal grouped-query attention. For token identifier t in layer l , a fixed table returns two experts. A seeded permutation of $t \bmod E$ determines the primary expert,

$$r_{l,1}(t) = \pi_l(t \bmod E), \quad (1)$$

and a table constructed in token-ID order assigns the least-used expert other than the primary as the secondary route. The assignment balances the number of vocabulary entries allocated to each expert, not their frequency in the training corpus.

For contextual hidden state $\mathbf{x}$, the feed-forward output is

$$\text{MLP}_l(\mathbf{x}, t) = g_l^s \text{Shared}_l(\mathbf{x}) + g_l^r \sum_{j=1}^2 0.5 \text{Expert}_{r_{l,j}(t)}(\mathbf{x}), \quad (2)$$

where g_l^s and g_l^r are learned scalar gates initialized to 1.0 and 0.1. Shared and routed branches are SwiGLU transformations. No auxiliary expert objective is used in the primary token-identity experiment.

The principal comparison controls the transformer backbone, tokenizer, data order, optimizer, schedule, sequence length, training-token budget and stored MLP width (Table 1). The routed model divides the dense width of 4,096 into a shared width of 3,840 and four stored experts of width 64.

Table 1: **Matched primary model configurations.** GQA, grouped-query attention. Stored routed width is $4 \times 64 = 256$; active routed width is $2 \times 64 = 128$.

Configuration	Token-identity residual	Dense
Transformer layers	18	18
Model dimension	1,024	1,024
Attention / key-value heads	16 / 4 (GQA)	16 / 4 (GQA)
Shared MLP width	3,840	—
Stored routed experts	4×64	—
Active routed experts	2	—
Dense MLP width	—	4,096
Vocabulary / context	32,000 / 2,048	32,000 / 2,048
Total parameters	306.5 million	306.5 million
Training budget	8 billion tokens	8 billion tokens

Table 2: **Matched-token loss trajectory.** All rows are training NLL except the explicitly labelled evaluation-stream row. Negative differences favour token routing. The evaluation stream is fixed across models but drawn from the FineWeb-Edu training split.

Measurement	Tokens	Dense	Token-routed	Difference
Train	105 million	6.6698	6.8464	+0.1766
Train	524 million	4.4450	4.4796	+0.0346
Train	1.05 billion	3.5500	3.5324	−0.0176
Train	2.10 billion	3.1953	3.1720	−0.0233
Train	4.19 billion	3.0925	3.0738	−0.0187
Train	6.29 billion	3.0061	2.9906	−0.0154
Evaluation stream	7.864 billion	2.9482	2.9329	−0.0153
Train	7.99 billion	2.9324	2.9231	−0.0093

With two active experts, its evaluated MLP width per token is 3,968. Both models contain 306.5 million parameters.

Token routing crosses dense after 0.8 billion tokens

The routed model begins behind the dense baseline and first crosses it at the logged training point near 0.78 billion tokens (Fig. 2). Its advantage then persists through the end of training. At the last common evaluation, after 7.864 billion tokens, evaluation-stream NLL is 2.9329 for token routing and 2.9482 for dense, a difference of -0.0153 (Table 2). Final logged training NLL is 2.9231 and 2.9324, respectively.

The fixed vocabulary assignment does not mathematically ensure balanced corpus traffic. Observed final traffic was 0.2481, 0.2635, 0.2481 and 0.2403 across the four experts, with no inactive expert. On the same eight-B300 system and global batch, dense training reached approximately 0.95 million tokens per second and token routing 0.75 million tokens per second. The quality comparison is consequently matched by tokens, not by wall-clock time or compute.

Standard evaluations show near parity

We evaluated the final step-7,629 checkpoints without task-specific fine-tuning using LM Evaluation Harness[5]. The token-routed model is 0.46 percentage points below dense on ARC-Easy[6], tied at displayed precision on PIQA[7], 0.39 points above dense on HellaSwag[8], and has 0.59 lower word

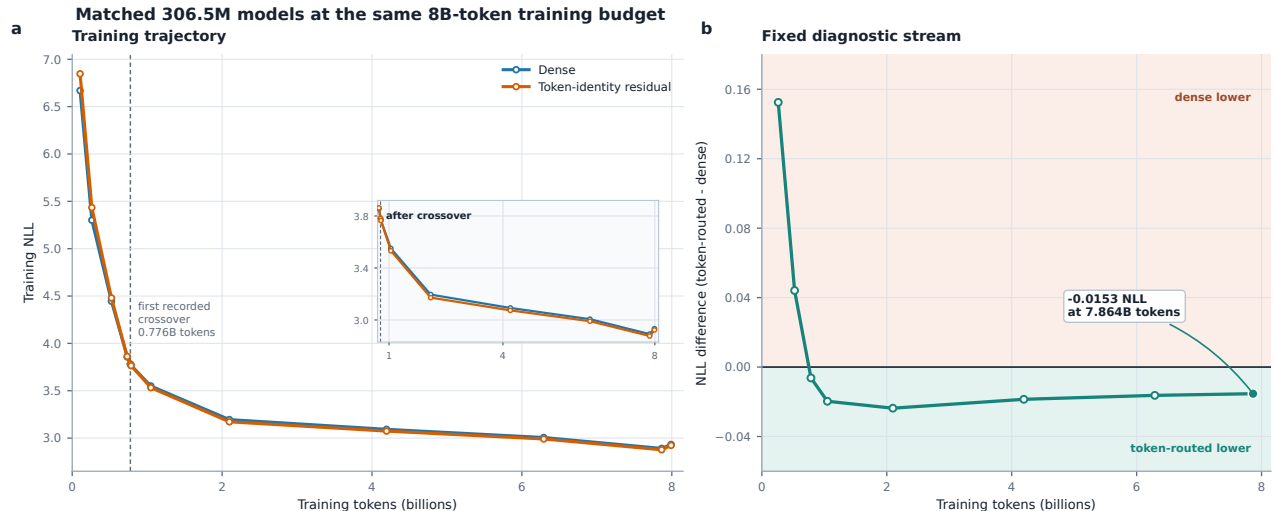


Figure 2: **Matched-token loss measurements.** **a**, Training NLL at recorded checkpoints, with the post-crossover region enlarged in the inset. The token-identity residual model (orange) begins behind dense (blue) and first crosses near 0.8 billion tokens. **b**, Token-identity residual minus dense NLL on the fixed FineWeb-Edu training-split diagnostic stream; negative values favour token routing. The final common evaluation differs by -0.0153 NLL after 7.864 billion tokens. Markers are recorded checkpoints and lines guide the eye. The diagnostic stream is not held out, and each curve represents one seed.

Table 3: **Zero-shot evaluation of the final primary checkpoints.** Multiple-choice values are length-normalized accuracy in percent, reported as estimate $\pm$ standard error. WikiText-2 reports word perplexity (lower is better). Sample counts are 2,376, 1,838 and 10,042 for ARC-Easy, PIQA and HellaSwag, respectively.

Model	ARC-Easy	PIQA	HellaSwag	WikiText-2
Dense	49.07 $\pm$ 1.03	63.49 $\pm$ 1.12	33.67 $\pm$ 0.47	35.79
Token-identity residual	48.61 $\pm$ 1.03	63.49 $\pm$ 1.12	34.06 $\pm$ 0.47	35.20

perplexity on WikiText-2[9] (Table 3). Each multiple-choice difference is smaller than one reported standard error. These tasks therefore support near parity, not task-level superiority.

Controls bound the interpretation

Three internally matched panels provide controls at smaller scale (Table 4). In the historical 1-billion-token panel, token-identity routing with a shared branch narrowly leads dense, whereas random routing and removing the shared branch are worse. In a separate 99.6-million-token diagnostic, an auxiliary-balanced learned top-2 router leads dense by 0.0119 evaluation NLL, while loss-free learned routing and fixed token-identity routing trail dense. Promoting the auxiliary router to a matched 1.0003-billion-token run reverses the short-budget ordering: dense leads by 0.0147 evaluation NLL and is approximately 5.2% faster. These panels use the same o200k tokenizer family but different data ingestion and evaluation streams from the historical controls, so their absolute losses are not compared.

Table 4: **Within-panel effects in smaller-scale controls.** Each value is the evaluation-NLL difference between a variant and the dense control from the same panel; negative values favour the variant. Panels use the same o200k tokenizer family, but the historical and contextual implementations use different data ingestion and evaluation streams.

Panel	Variant versus matched dense	Tokens	Δ evaluation NLL
Historical fixed	Shared + balanced secondary	1.0003B	−0.0071
	Shared + random top-2	1.0003B	+0.0619
	Balanced secondary, no shared	1.0003B	+0.1435
Contextual short	Learned top-2 + auxiliary balancing	99.6M	−0.0119
	Learned top-2 + loss-free balancing	99.6M	+0.0286
	Fixed token-identity top-2	99.6M	+0.0507
Contextual full	Learned top-2 + auxiliary balancing	1.0003B	+0.0147

Discussion

These experiments isolate a modest effect: a fixed token identifier can select a useful residual parameter subspace while contextual computation remains shared. The shared-context/token-identity separation is therefore an architectural building block, not a demonstrated replacement for the transformer backbone. It could in principle be paired with another contextual substrate, but that possibility requires direct experiments. The result does not imply that token identifiers encode semantic expert classes. The primary assignments are modulo partitions up to a layer-specific permutation, and every active expert transforms the contextual hidden state. Stable token-conditioned access to parameters is sufficient to explain the observation.

The comparison also exposes practical trade-offs. Stored parameter count is matched, but active width differs slightly and the routed implementation is 21% slower at the measured batch. Token routing therefore offers no demonstrated wall-clock or hardware-efficiency advantage. The auxiliary-balanced learned router leads dense at short budget but trails it in loss and throughput at the full budget. These single-seed controls do not establish that fixed routing is generally preferable.

The largest limitations are statistical and evaluative. Each architecture has one training seed, the loss-evaluation streams come from the training split, and the standard task panel is small. Accuracy gaps remain below one standard error. Replication across seeds, an independent held-out corpus, stronger out-of-distribution tasks and larger models are needed to assess robustness. A matched compute or wall-clock study would also be required to evaluate efficiency. The present evidence should consequently be read as a controlled architectural finding and a hypothesis for broader MoE studies.

Fixed routing removes one adaptive component and can simplify analysis, but it can also concentrate recurring token types in stable parameter subsets. We did not evaluate memorization, demographic performance, misuse or deployment safety. Any release of weights or application of the method should be accompanied by task-appropriate safety and data-governance evaluation.

Methods

Transformer and residual experts

Both primary models are decoder-only transformers with 18 layers, model dimension 1,024, 16 attention heads, four key-value heads, rotary positional embeddings, QK normalization, causal scaled dot-product attention and a maximum sequence length of 2,048. Feed-forward transformations use SwiGLU,

$$\text{SwiGLU}(\mathbf{x}) = (\text{SiLU}(\mathbf{x}\mathbf{W}_{\text{gate}}) \odot \mathbf{x}\mathbf{W}_{\text{up}})\mathbf{W}_{\text{down}}. \quad (3)$$

For model dimension d , shared width d_s , expert width d_e , E stored experts and k active experts, the routed layer stores $3d(d_s + Ed_e)$ weights and evaluates intermediate width $d_s + kd_e$. Tokens are grouped by their fixed assignments so that only selected routed experts are evaluated. The two selected outputs receive fixed weights 0.5 and 0.5 before multiplication by the learned routed gate.

Construction and verification of fixed routes

Each layer uses a seeded permutation of four expert indices. Primary routes are obtained from the permuted token-ID residue. The secondary lookup is constructed once by iterating over token identifiers in ascending order and selecting, with lowest-index tie breaking, the least-used expert other than the primary. With a 32,000-entry vocabulary, every expert receives 8,000 primary identifiers. Checkpoint inspection verified that lookup buffers in all 18 layers reproduce this rule exactly. The construction uses no token counts or frequency table.

Primary training protocol

Both primary models were trained on the same FineWeb-Edu token stream for 7,629 steps, corresponding to approximately 8 billion tokens. The tokenizer has 32,000 entries and the global batch contains 1,048,576 tokens (eight GPUs, batch 64 per GPU, sequence length 2,048). Training uses BF16 and AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1 and gradient clipping at 1.0. The learning rate uses a 5% warm-up followed by cosine decay to 10% of the peak. Residual output projections are initialized with standard deviation $0.02/\sqrt{2L}$ for L layers[10]. Apart from the MLP parameterization, run configuration, initialization seed, data order and token budget are matched.

Training NLL is read directly from the run metrics at matched steps. The fixed evaluation stream is identical for both models but sampled from the FineWeb-Edu training split; it is therefore a diagnostic of training progress, not held-out generalization. We report the last common evaluation point rather than an average of the final 50 training measurements. Throughput is the recorded steady-state training throughput on the same eight-NVIDIA-B300 system.

Control experiments

The historical fixed-routing panel consists of seven approximately 100-million-parameter models trained for 1.0003 billion tokens on two NVIDIA B200 GPUs. It tests a dense residual MLP, a shared-only branch, a routed-only branch, random assignment and three deterministic top-2 labels. Inspection showed that the condition historically configured as frequency-aware had no populated frequency table and realized modulo-primary/cardinality-balanced-secondary assignment. Some named controls have equivalent primary assignments; differences among them are descriptive.

The contextual-router panels use the same o200k tokenizer family, a separate FineWeb-Edu memory-mapped shard, two NVIDIA RTX PRO 6000 Blackwell GPUs and a 98.2-million-parameter backbone. The learned router scores four residual experts from hidden states and selects two. The 99.6-million-token diagnostic compares dense, fixed token-identity routing, auxiliary-balanced learned routing and loss-free learned routing with the same seed, token order, batch and schedule. Only the auxiliary variant is promoted to a matched 1.0003-billion-token run against dense. Reported training NLL excludes the auxiliary term. The 1-billion-token runs have a 954-step budget, log training metrics every ten steps and evaluate every 750 steps; step 950 is therefore the last logged training point and step 750 the last evaluation checkpoint. The short runs analogously report training at step 95 and evaluation at step 75.

Standard evaluation

Final step-7,629 checkpoints were exported to the same BF16 inference runtime. A five-prompt paired generation sanity test was completed before task evaluation. EleutherAI LM Evaluation Harness version 0.4.12 ran ARC-Easy, PIQA, HellaSwag and WikiText-2 with no sample limit, zero-shot prompts, automatic batching, a 2,048-token maximum context and seeds 1234 for Python, NumPy, PyTorch and few-shot sampling. We report `acc_norm` for multiple-choice tasks and `word_perplexity` for WikiText-2. The runtime used Transformers 4.57.6 and Datasets 5.0.0.

Statistics and reproducibility

Training experiments use one seed per architecture; no confidence interval or hypothesis test is reported for loss. Multiple-choice standard errors are the task-level standard errors returned by the evaluation harness. Differences are shown at the precision supported by the measurements and are not converted into claims of statistical significance. Source metrics, configurations, route-verification tests, figure scripts and aggregate benchmark records accompany the supplementary code.

Use of generative artificial intelligence

Generative artificial-intelligence tools were used as general-purpose assistants for language editing, consistency checks between the manuscript and author-provided CSV/JSON metric files, and figure and revision scripting. The authors designed and executed the experiments, reviewed and verified all experimental results, numerical values, scientific claims and final text, and take full responsibility for the manuscript.

Data availability

FineWeb-Edu is publicly available under its applicable repository terms. The exact tokenizers, preprocessing instructions, run configurations and aggregate results needed to reproduce the reported analyses are included in the anonymized peer-review archive. The full checkpoint files and per-example evaluation logs are available to editors and reviewers on request and will be deposited in a persistent public repository upon publication.

Code availability

An anonymized source archive containing the PyTorch implementation, fixed and learned routing variants, route tests, training entry points and analysis scripts accompanies the submission. A public, versioned repository and archival release will be provided upon publication.

Author contributions

Author names and contribution details are withheld from this double-anonymized manuscript and are supplied in the cover letter.

Competing interests

A complete competing-interests declaration is supplied in the cover letter for double-anonymized review.

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